```python
def update_parameters_with_gd(parameters, grads, learning_rate):
    """
    Update parameters using one step of gradient descent

    Arguments:
    parameters -- python dictionary containing your parameters to be updated:
                    parameters['W' + str(l)] = Wl
                    parameters['b' + str(l)] = bl
    grads -- python dictionary containing your gradients to update each parameters:
                    grads['dW' + str(l)] = dWl
                    grads['db' + str(l)] = dbl
    learning_rate -- the learning rate, scalar.

    Returns:
    parameters -- python dictionary containing your updated parameters
    """

    L = len(parameters) // 2 # number of layers in the neural networks
    # Update rule for each parameter
    for l in range(L):
        ### START CODE HERE ### (approx. 2 lines)
        parameters["W" + str(l+1)] = parameters['W'+str(l+1)]-learning_rate*grads["dW"+str(l+1)]
        parameters["b" + str(l+1)] = parameters['b'+str(l+1)]-learning_rate*grads["db"+str(l+1)]
        ### END CODE HERE ###

    return parameters
```

```
W1 = [[ 1.63535156 -0.62320365 -0.53718766]
 [-1.07799357  0.85639907 -2.29470142]]
b1 = [[ 1.74604067]
 [-0.75184921]]
W2 = [[ 0.32171798 -0.25467393  1.46902454]
 [-2.05617317 -0.31554548 -0.3756023 ]
 [ 1.1404819  -1.09976462 -0.1612551 ]]
b2 = [[-0.88020257]
 [ 0.02561572]
 [ 0.57539477]]
```

```python
def random_mini_batches(X, Y, mini_batch_size = 64, seed = 0):
    """
    Creates a list of random minibatches from (X, Y)

    Arguments:
    X -- input data, of shape (input size, number of examples)
    Y -- true "label" vector (1 for blue dot / 0 for red dot), of shape (1, number of examples)
    mini_batch_size -- size of the mini-batches, integer

    Returns:
    mini_batches -- list of synchronous (mini_batch_X, mini_batch_Y)
    """

    np.random.seed(seed)            # To make your "random" minibatches the same as ours
    m = X.shape[1]                  # number of training examples
    mini_batches = []

    # Step 1: Shuffle (X, Y)
    permutation = list(np.random.permutation(m))
    shuffled_X = X[:, permutation]
    shuffled_Y = Y[:, permutation].reshape((1,m))

    # Step 2: Partition (shuffled_X, shuffled_Y). Minus the end case.
    num_complete_minibatches = math.floor(m/mini_batch_size) # number of mini batches of size mini_
    for k in range(0, num_complete_minibatches):
        ### START CODE HERE ### (approx. 2 lines)
        mini_batch_X = shuffled_X[:,mini_batch_size*k:mini_batch_size*(k+1)]
        mini_batch_Y = shuffled_Y[:,mini_batch_size*k:mini_batch_size*(k+1)]
        ### END CODE HERE ###
        mini_batch = (mini_batch_X, mini_batch_Y)
        mini_batches.append(mini_batch)

    # Handling the end case (last mini-batch < mini_batch_size)
    if m % mini_batch_size != 0:
        ### START CODE HERE ### (approx. 2 lines)
        mini_batch_X = shuffled_X[:,mini_batch_size*num_complete_minibatches:]
        mini_batch_Y = shuffled_Y[:,mini_batch_size*num_complete_minibatches:]
        ### END CODE HERE ###
        mini_batch = (mini_batch_X, mini_batch_Y)
        mini_batches.append(mini_batch)

    return mini_batches
```

```
shape of the 1st mini_batch_X: (12288, 64)
shape of the 2nd mini_batch_X: (12288, 64)
shape of the 3rd mini_batch_X: (12288, 20)
shape of the 1st mini_batch_Y: (1, 64)
shape of the 2nd mini_batch_Y: (1, 64)
shape of the 3rd mini_batch_Y: (1, 20)
mini batch sanity check: [ 0.90085595 -0.7612069   0.2344157 ]
```

利用 np.random.permutation(m)創造出 1*m 的隨機數列矩陣當作 shuffle 的位置。這些位置將要使用 shuffle：

```
[24, 7, 37, 114, 27, 26, 43, 54, 132, 71, 76, 22, 83, 130, 100, 73, 44, 16, 51, 66, 8, 133, 93, 90, 86, 40, 101, 107, 92, 145,
33, 78, 18, 127, 63, 45, 2, 85, 10, 125, 109, 62, 112, 122, 102, 61, 139, 116, 120, 96, 50, 60, 56, 30, 59, 135, 91, 121, 84, 9
4, 118, 48, 142, 131, 13, 104, 119, 20, 15, 52, 3, 123, 105, 6, 68, 89, 12, 95, 113, 97, 46, 11, 144, 108, 41, 110, 1, 106, 13
7, 42, 4, 124, 17, 38, 5, 53, 141, 98, 0, 34, 28, 55, 75, 35, 23, 74, 31, 111, 57, 129, 65, 32, 136, 14, 146, 19, 29, 126, 49,
134, 99, 82, 64, 147, 79, 69, 128, 80, 115, 143, 72, 77, 25, 81, 138, 140, 39, 58, 88, 70, 87, 36, 21, 9, 103, 67, 117, 47]
```

mini_batch_X = shuffled_X[:,mini_batch_size*k:mini_batch_size*(k+1)]
將 shuffle 好的 X 分割， “mini_batch_size*k:mini_batch_size*(k+1)”表分割從第 64*k 加 64 項，最後剩下的再用”mini_batch_size*num_complete_minibatches:”表從最後一個 64*k 到最後一項。

```python
def initialize_velocity(parameters):
    """
    Initializes the velocity as a python dictionary with:
                - keys: "dW1", "db1", ..., "dWL", "dbL"
                - values: numpy arrays of zeros of the same shape as the correspondi
    Arguments:
    parameters -- python dictionary containing your parameters.
                    parameters['W' + str(l)] = Wl
                    parameters['b' + str(l)] = bl

    Returns:
    v -- python dictionary containing the current velocity.
                    v['dW' + str(l)] = velocity of dWl
                    v['db' + str(l)] = velocity of dbl
    """

    L = len(parameters) // 2 # number of layers in the neural networks
    v = {}

    # Initialize velocity
    for l in range(L):
        ### START CODE HERE ### (approx. 2 lines)
        v["dW" + str(l+1)] = np.zeros(parameters['W'+str(l+1)].shape)
        v["db" + str(l+1)] = np.zeros(parameters['b'+str(l+1)].shape)
        ### END CODE HERE ###

    return v
```

```
v["dW1"] = [[0. 0. 0.]
 [0. 0. 0.]]
v["db1"] = [[0.]
 [0.]]
v["dW2"] = [[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
v["db2"] = [[0.]
 [0.]
 [0.]]
```

```python
def update_parameters_with_momentum(parameters, grads, v, beta, learning_rate):
    """
    Update parameters using Momentum

    Arguments:
    parameters -- python dictionary containing your parameters:
                    parameters['W' + str(l)] = Wl
                    parameters['b' + str(l)] = bl
    grads -- python dictionary containing your gradients for each parameters:
                    grads['dW' + str(l)] = dWl
                    grads['db' + str(l)] = dbl
    v -- python dictionary containing the current velocity:
                    v['dW' + str(l)] = ...
                    v['db' + str(l)] = ...
    beta -- the momentum hyperparameter, scalar
    learning_rate -- the learning rate, scalar

    Returns:
    parameters -- python dictionary containing your updated parameters
    v -- python dictionary containing your updated velocities
    """

    L = len(parameters) // 2 # number of layers in the neural networks

    # Momentum update for each parameter
    for l in range(L):

        ### START CODE HERE ### (approx. 4 lines)
        # compute velocities
        v["dW" + str(l+1)] = beta*v["dW" + str(l+1)]+(1-beta)*grads["dW" + str(l+1)]
        v["db" + str(l+1)] = beta*v["db" + str(l+1)]+(1-beta)*grads["db" + str(l+1)]
        # update parameters
        parameters["W" + str(l+1)] =  parameters["W" + str(l+1)]-learning_rate*v["dW" + str(l+1)]
        parameters["b" + str(l+1)] = parameters["b" + str(l+1)]-learning_rate*v["db" + str(l+1)]
        ### END CODE HERE ###

    return parameters, v
```

```
W1 = [[ 1.62544598 -0.61290114 -0.52907334]
 [-1.07347112  0.86450677 -2.30085497]]
b1 = [[ 1.74493465]
 [-0.76027113]]
W2 = [[ 0.31930698 -0.24990073  1.4627996 ]
 [-2.05974396 -0.32173003 -0.38320915]
 [ 1.13444069 -1.0998786  -0.1713109 ]]
b2 = [[-0.87809283]
 [ 0.04055394]
 [ 0.58207317]]
v["dW1"] = [[-0.11006192  0.11447237  0.09015907]
 [ 0.05024943  0.09008559 -0.06837279]]
v["db1"] = [[-0.01228902]
 [-0.09357694]]
v["dW2"] = [[-0.02678881  0.05303555 -0.06916608]
 [-0.03967535 -0.06871727 -0.08452056]
 [-0.06712461 -0.00126646 -0.11173103]]
v["db2"] = [[0.02344157]
 [0.16598022]
 [0.07420442]]
```

```
def initialize_adam(parameters) :
    """
    Initializes v and s as two python dictionaries with:
                - keys: "dW1", "db1", ..., "dWL", "dbL"
                - values: numpy arrays of zeros of the same shape as the corre

    Arguments:
    parameters -- python dictionary containing your parameters.
                    parameters["W" + str(l)] = Wl
                    parameters["b" + str(l)] = bl

    Returns:
    v -- python dictionary that will contain the exponentially weighted averag
                    v["dW" + str(l)] = ...
                    v["db" + str(l)] = ...
    s -- python dictionary that will contain the exponentially weighted averag
                    s["dW" + str(l)] = ...
                    s["db" + str(l)] = ...

    """

    L = len(parameters) // 2 # number of layers in the neural networks
    v = {}
    s = {}

    # Initialize v, s. Input: "parameters". Outputs: "v, s".
    for l in range(L):
    ### START CODE HERE ### (approx. 4 lines)
        v["dW" + str(l+1)] = np.zeros(parameters['W'+str(l+1)].shape)
        v["db" + str(l+1)] = np.zeros(parameters['b'+str(l+1)].shape)
        s["dW" + str(l+1)] = np.zeros(parameters['W'+str(l+1)].shape)
        s["db" + str(l+1)] = np.zeros(parameters['b'+str(l+1)].shape)
    ### END CODE HERE ###

    return v, s
```

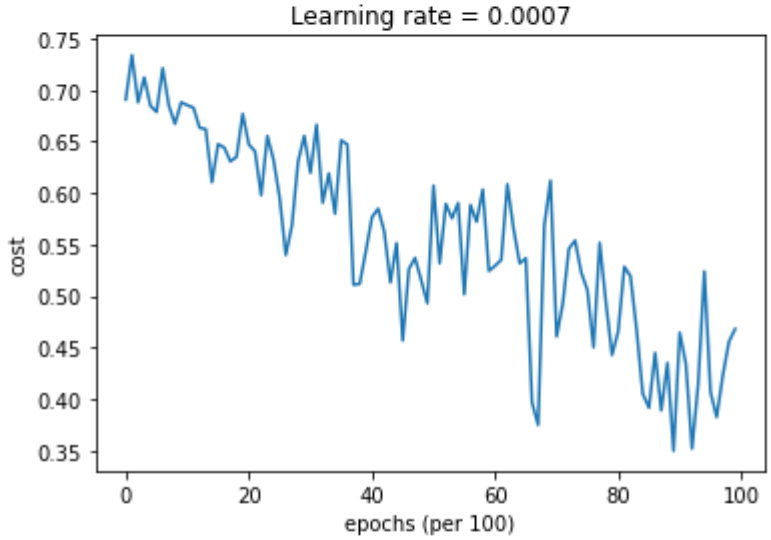
```
v["dW1"] = [[0. 0. 0.]
 [0. 0. 0.]]
v["db1"] = [[0.]
 [0.]]
v["dW2"] = [[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
v["db2"] = [[0.]
 [0.]
 [0.]]
s["dW1"] = [[0. 0. 0.]
 [0. 0. 0.]]
s["db1"] = [[0.]
 [0.]]
s["dW2"] = [[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
s["db2"] = [[0.]
 [0.]
 [0.]]
```

```python
def update_parameters_with_adam(parameters, grads, v, s, t, learning_rate = 0.01,
                                beta1 = 0.9, beta2 = 0.999,  epsilon = 1e-8):

    L = len(parameters) // 2                 # number of layers in the neural networks
    v_corrected = {}                         # Initializing first moment estimate, python dictionary
    s_corrected = {}                         # Initializing second moment estimate, python dictionary

    # Perform Adam update on all parameters
    for l in range(L):
        # Moving average of the gradients. Inputs: "v, grads, beta1". Output: "v".
        ### START CODE HERE ### (approx. 2 lines)
        v["dW" + str(l+1)] = beta1*v["dW"+str(l+1)]+(1-beta1)*grads['dW'+str(l+1)]
        v["db" + str(l+1)] = beta1*v["db"+str(l+1)]+(1-beta1)*grads['db'+str(l+1)]
        ### END CODE HERE ###

        # Compute bias-corrected first moment estimate. Inputs: "v, beta1, t". Output: "v_corrected".
        ### START CODE HERE ### (approx. 2 lines)
        v_corrected["dW" + str(l+1)] = v["dW" + str(l+1)]/(1-beta1**t)
        v_corrected["db" + str(l+1)] = v["db" + str(l+1)]/(1-beta1**t)
        ### END CODE HERE ###

        # Moving average of the squared gradients. Inputs: "s, grads, beta2". Output: "s".
        ### START CODE HERE ### (approx. 2 lines)
        s["dW" + str(l+1)] = beta2*s["dW"+str(l+1)]+(1-beta2)*(grads['dW'+str(l+1)]**2)
        s["db" + str(l+1)] = beta2*s["db"+str(l+1)]+(1-beta2)*(grads['db'+str(l+1)]**2)
        ### END CODE HERE ###

        # Compute bias-corrected second raw moment estimate. Inputs: "s, beta2, t". Output: "s_corrected".
        ### START CODE HERE ### (approx. 2 lines)
        s_corrected["dW" + str(l+1)] = s["dW" + str(l+1)]/(1-beta2**t)
        s_corrected["db" + str(l+1)] = s["db" + str(l+1)]/(1-beta2**t)
        ### END CODE HERE ###

        # Update parameters. Inputs: "parameters, learning_rate, v_corrected, s_corrected, epsilon". Output: "parameters".
        ### START CODE HERE ### (approx. 2 lines)
        parameters["W" + str(l+1)] = parameters["W" + str(l+1)]-learning_rate*(
            (v_corrected["dW" + str(l+1)])/(np.sqrt(s_corrected["dW" + str(l+1)])+epsilon))
        parameters["b" + str(l+1)] = parameters["b" + str(l+1)]-learning_rate*(
            (v_corrected["db" + str(l+1)])/(np.sqrt(s_corrected["db" + str(l+1)])+epsilon))
        ### END CODE HERE ###

    return parameters, v, s
```

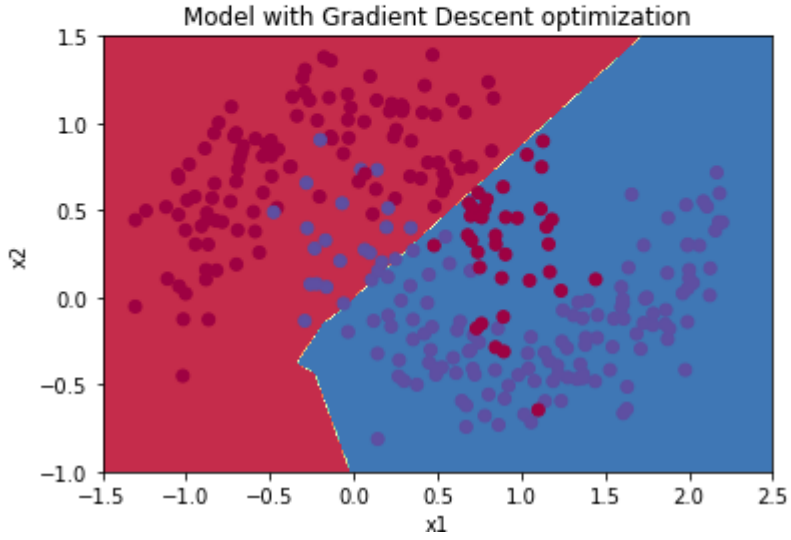
```
W1 = [[ 1.63178673 -0.61919778 -0.53561312]
 [-1.08040999  0.85796626 -2.29409733]]
b1 = [[ 1.75225313]
 [-0.75376553]]
W2 = [[ 0.32648046 -0.25681174  1.46954931]
 [-2.05269934 -0.31497584 -0.37661299]
 [ 1.14121081 -1.09244991 -0.16498684]]
b2 = [[-0.88529979]
 [ 0.03477238]
 [ 0.57537385]]
v["dW1"] = [[-0.11006192  0.11447237  0.09015907]
 [ 0.05024943  0.09008559 -0.06837279]]
v["db1"] = [[-0.01228902]
 [-0.09357694]]
v["dW2"] = [[-0.02678881  0.05303555 -0.06916608]
 [-0.03967535 -0.06871727 -0.08452056]
 [-0.06712461 -0.00126646 -0.11173103]]
v["db2"] = [[0.02344157]
 [0.16598022]
 [0.07420442]]
s["dW1"] = [[0.00121136 0.00131039 0.00081287]
 [0.0002525  0.00081154 0.00046748]]
s["db1"] = [[1.51020075e-05]
 [8.75664434e-04]]
s["dW2"] = [[7.17640232e-05 2.81276921e-04 4.78394595e-04]
 [1.57413361e-04 4.72206320e-04 7.14372576e-04]
 [4.50571368e-04 1.60392066e-07 1.24838242e-03]]
s["db2"] = [[5.49507194e-05]
 [2.75494327e-03]
 [5.50629536e-04]]
```

將 epoch 設為 10000 次：

```
Cost after epoch 0: 0.690736
Cost after epoch 1000: 0.685273
Cost after epoch 2000: 0.647072
Cost after epoch 3000: 0.619525
Cost after epoch 4000: 0.576584
Cost after epoch 5000: 0.607243
Cost after epoch 6000: 0.529403
Cost after epoch 7000: 0.460768
Cost after epoch 8000: 0.465586
Cost after epoch 9000: 0.464518
```

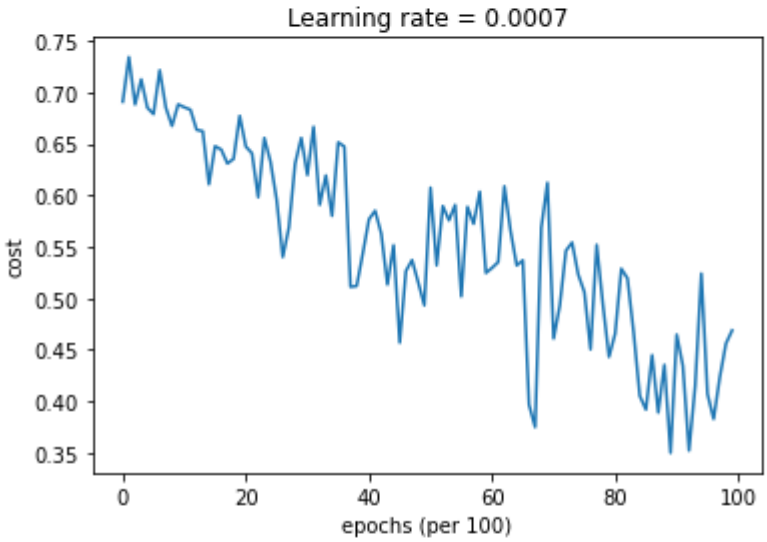



```
Accuracy: 0.7966666666666666
```

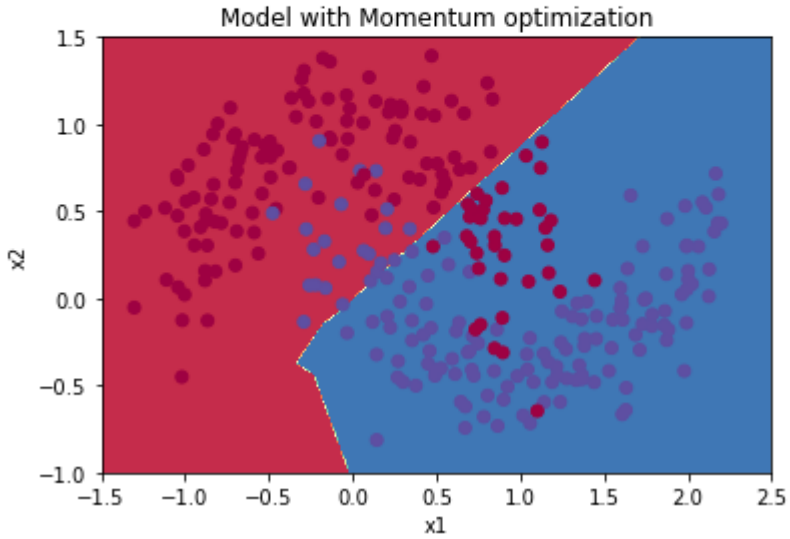


(GD)

```
Cost after epoch 0: 0.690741
Cost after epoch 1000: 0.685341
Cost after epoch 2000: 0.647145
Cost after epoch 3000: 0.619594
Cost after epoch 4000: 0.576665
Cost after epoch 5000: 0.607324
Cost after epoch 6000: 0.529476
Cost after epoch 7000: 0.460936
Cost after epoch 8000: 0.465780
Cost after epoch 9000: 0.464740
```

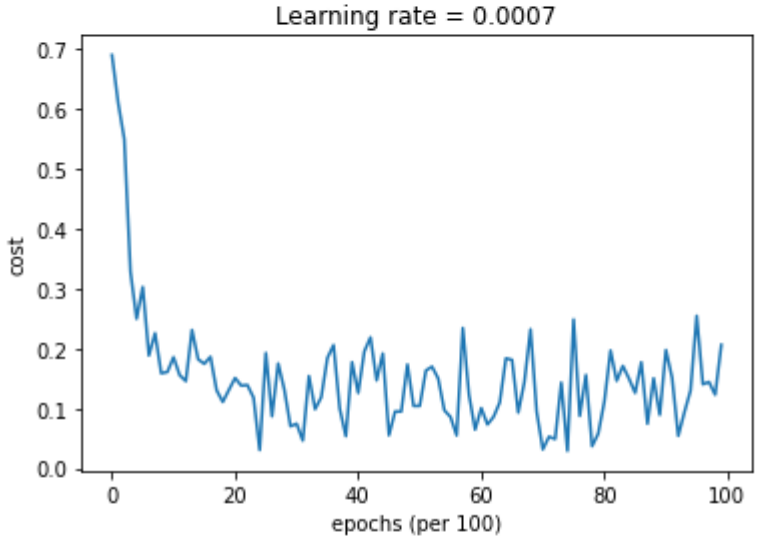



```
Accuracy: 0.7966666666666666
```



(Mom)

```
Cost after epoch 0: 0.690552
Cost after epoch 1000: 0.185567
Cost after epoch 2000: 0.150852
Cost after epoch 3000: 0.074454
Cost after epoch 4000: 0.125936
Cost after epoch 5000: 0.104235
Cost after epoch 6000: 0.100552
Cost after epoch 7000: 0.031601
Cost after epoch 8000: 0.111709
Cost after epoch 9000: 0.197648
```



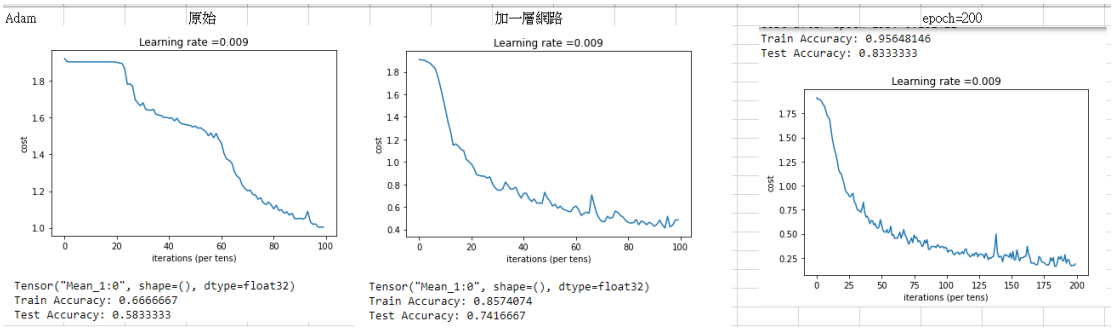

```
Accuracy: 0.94
```

(Adam)

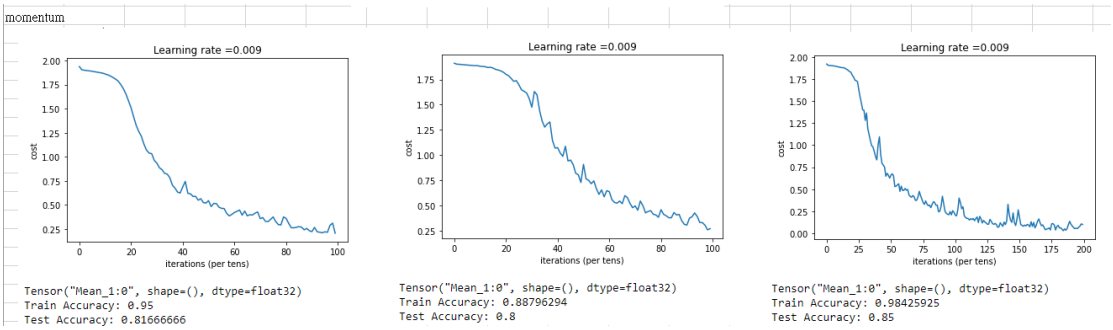
可以看到 Adam 擬合的效果最好，Adam 的數學式考慮了更多情況，較不容受到 local mim 的影響。

討論上次作業 HW6

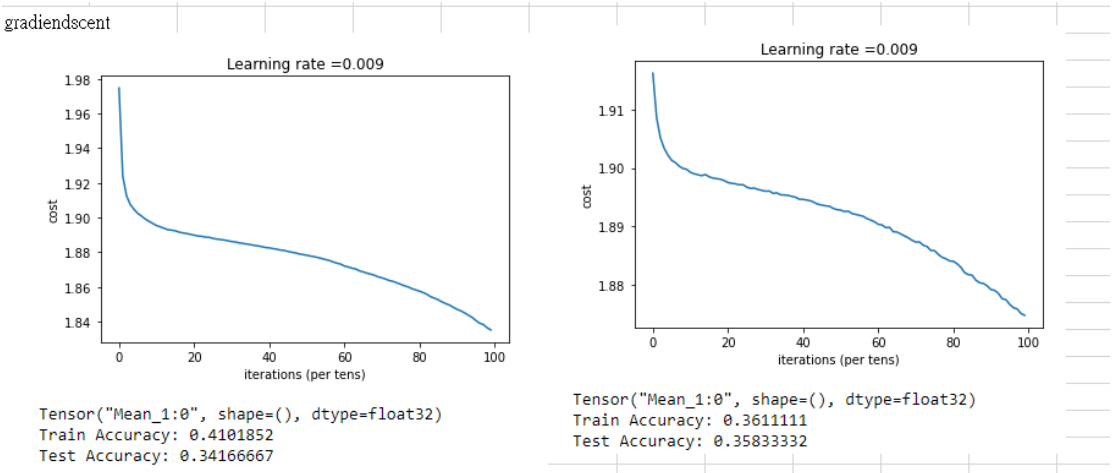
Adam



Mom



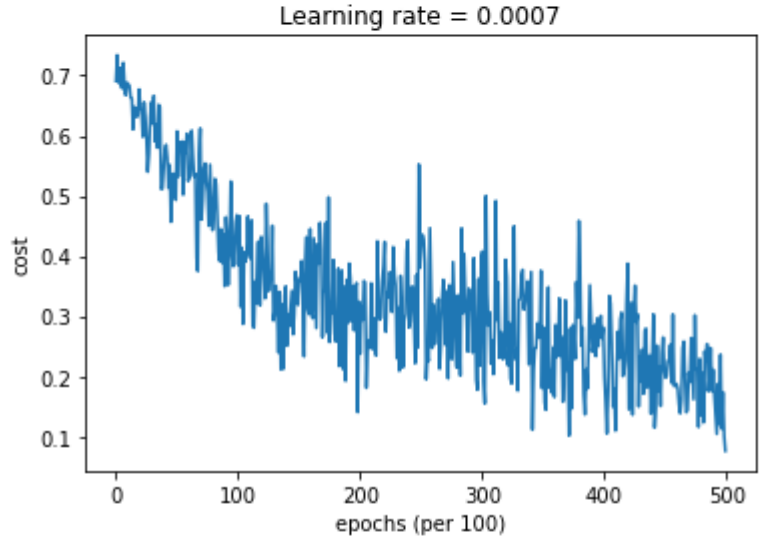
GD



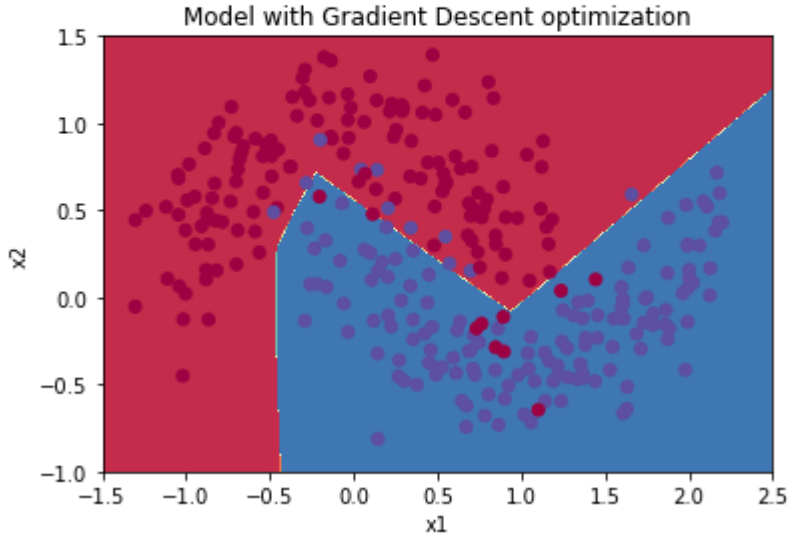
由上次的作業能看到 GD 處理影像並不理想，可能受限於 local mim 或訓練次數不夠多，對於 Adam 和 momentum 而言，momentum 訓練速度比 Adam 快上許多，準確率亦不輸 Adam，考慮到動能的概念，兩者比起 GD 較不容易受 local mim 影響。

現將 epoch 提高到 50000 次

```
Cost after epoch 40000: 0.198111
Cost after epoch 41000: 0.217824
Cost after epoch 42000: 0.144920
Cost after epoch 43000: 0.191010
Cost after epoch 44000: 0.303087
Cost after epoch 45000: 0.197643
Cost after epoch 46000: 0.179658
Cost after epoch 47000: 0.190288
Cost after epoch 48000: 0.205384
Cost after epoch 49000: 0.168115
```

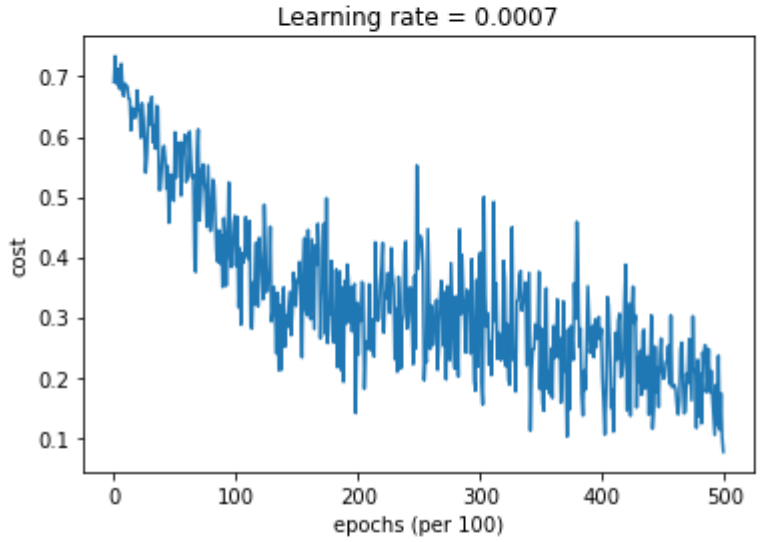



```
Accuracy: 0.9333333333333333
```

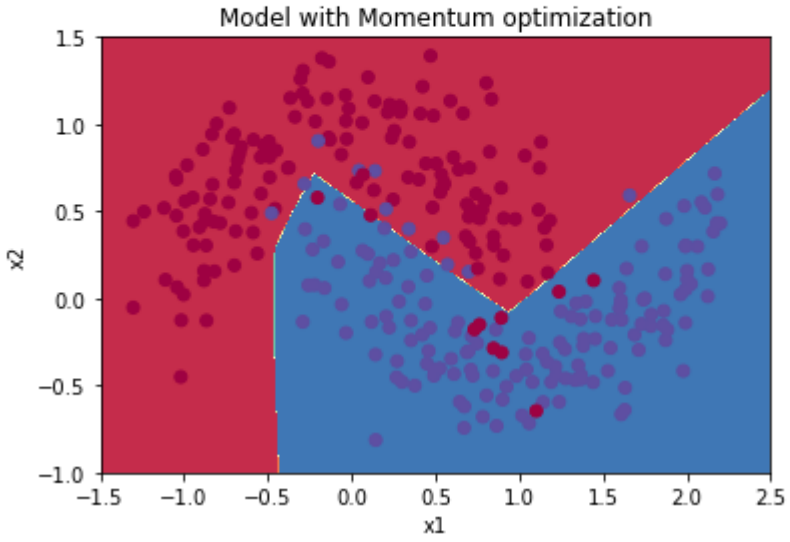


(GD)

```
Cost after epoch 40000: 0.198086
Cost after epoch 41000: 0.217775
Cost after epoch 42000: 0.144886
Cost after epoch 43000: 0.190995
Cost after epoch 44000: 0.303073
Cost after epoch 45000: 0.197581
Cost after epoch 46000: 0.179661
Cost after epoch 47000: 0.190263
Cost after epoch 48000: 0.205303
Cost after epoch 49000: 0.168090
```

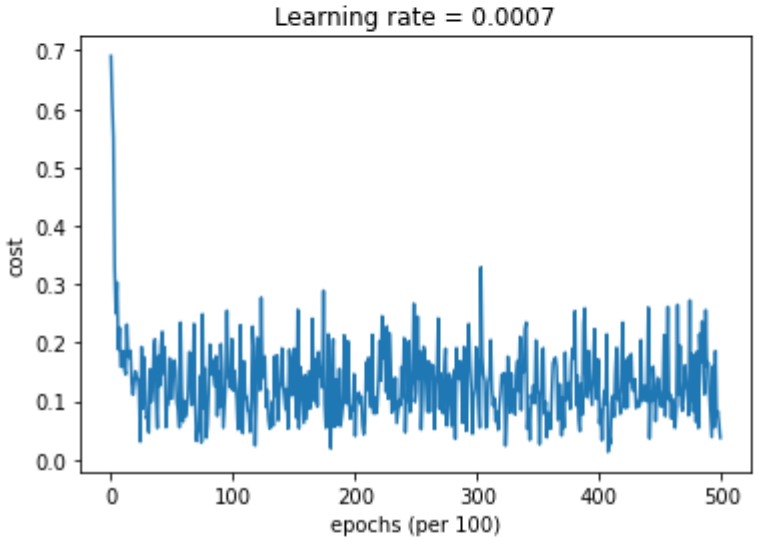



```
Accuracy: 0.9333333333333333
```



(Mom)

```
Cost after epoch 40000: 0.061768
Cost after epoch 41000: 0.097047
Cost after epoch 42000: 0.116761
Cost after epoch 43000: 0.087885
Cost after epoch 44000: 0.260130
Cost after epoch 45000: 0.092043
Cost after epoch 46000: 0.102810
Cost after epoch 47000: 0.161086
Cost after epoch 48000: 0.184762
Cost after epoch 49000: 0.115083
```

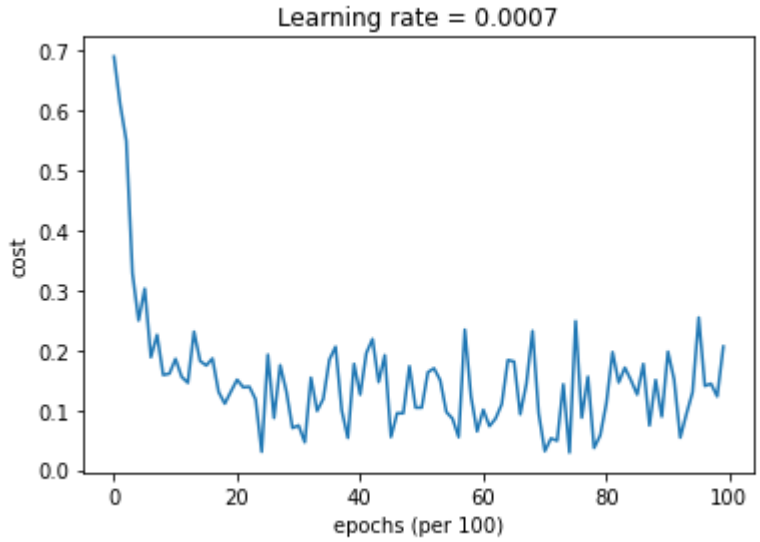



```
Accuracy: 0.9433333333333334
```

(Adam)

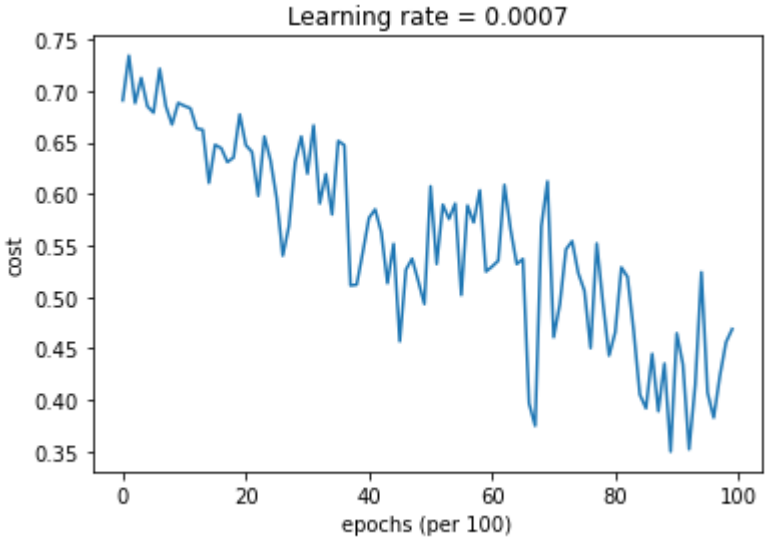
由三組分布圖可以看到擬合狀況都差不多，因此正確率也應為差不多，在訓練後期(>30000 epoch)可以看到 Adam 震盪其值收斂，GD 和 momentum 還有下降空間，再加 epoch 應能再提高正確率。

現將 minibatch 縮小，epoch 設為 10000

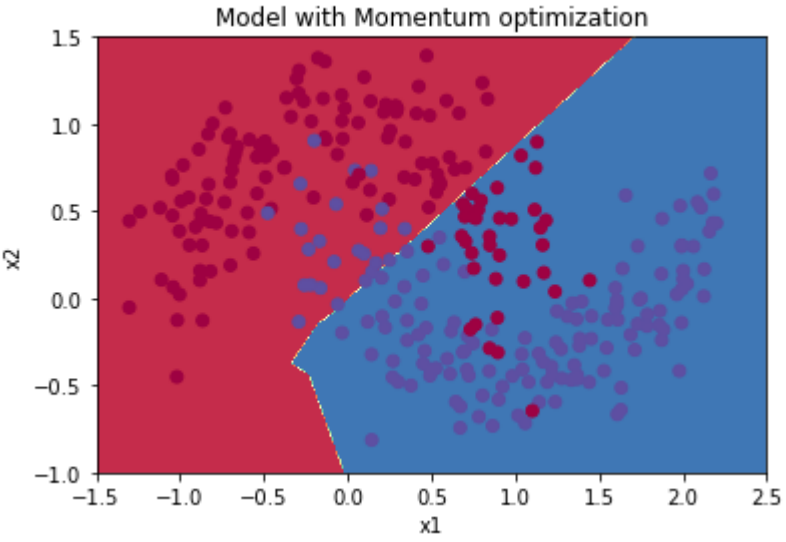



```
Accuracy: 0.94
```

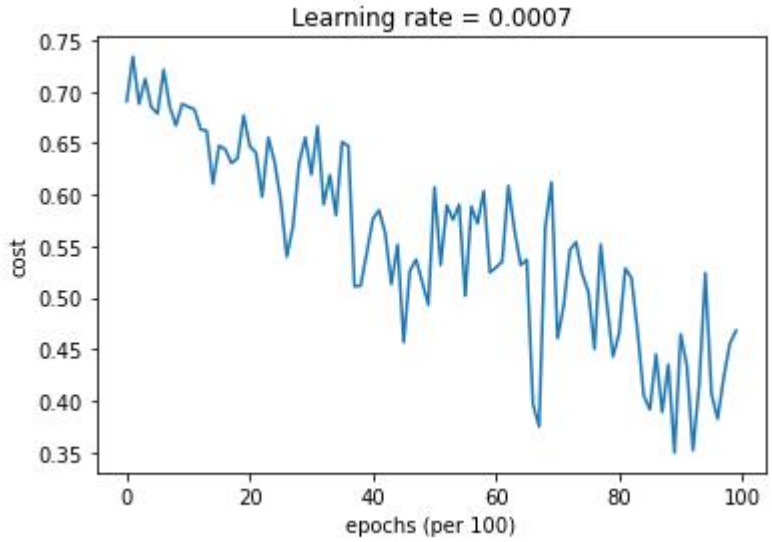
(Adam)



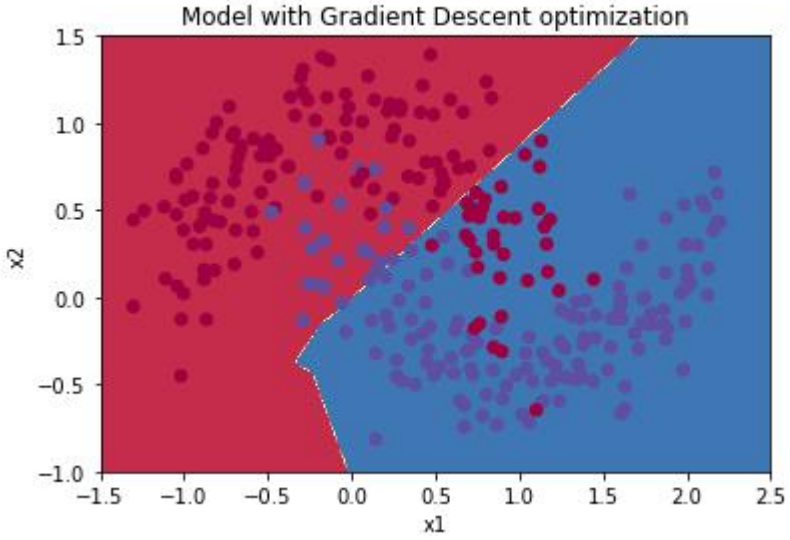

```
Accuracy: 0.7966666666666666
```



(Mom)



Accuracy: 0.7966666666666666



(GD)

很明顯看到正確率與 minibatch=64 相同。將 minibatch 調整為 32、16 結果亦同，推測 minibatch 方法在簡單且數目小的數據影響並不大。